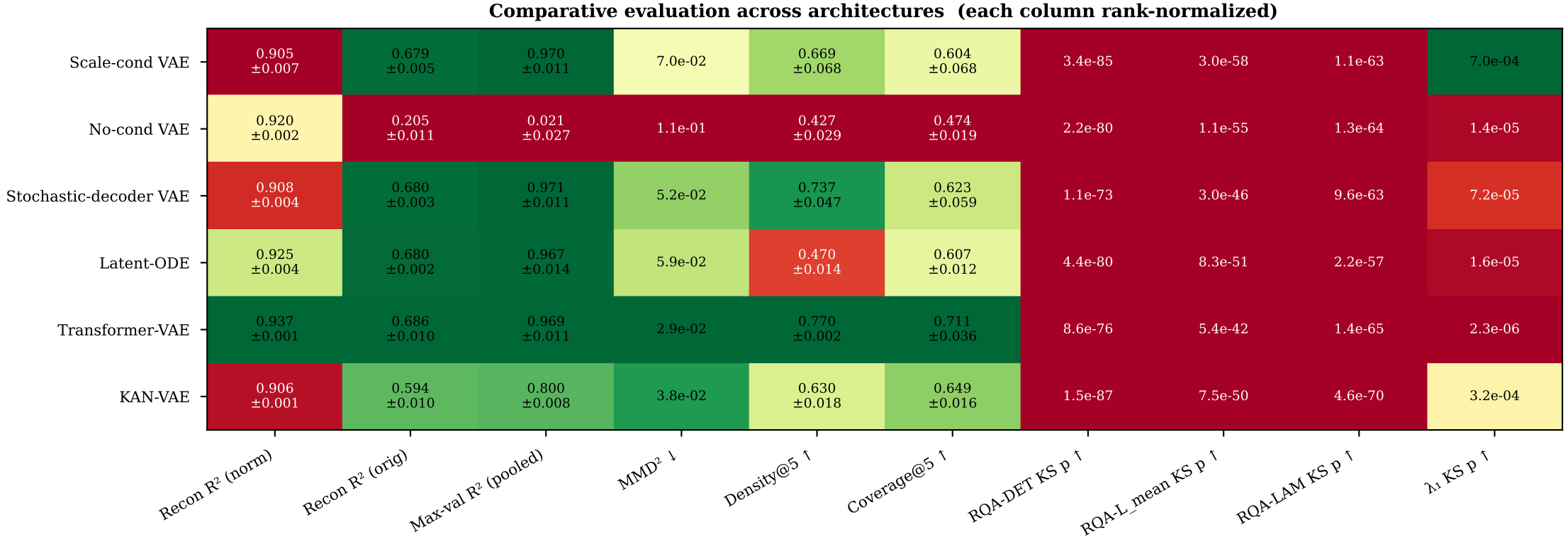
Comparative evaluation across architectures (each column rank-normalized)
Scale-cond VAE
No-cond VAE
Stochastic-decoder VAE
Latent-ODE
Transformer-VAE
KAN-VAE
0.905 ±0.007
0.679 ±0.005
0.970 ±0.011
7.0e-02
0.669 ±0.068
0.604 ±0.068
3.4e-85
3.0e-58
1.1e-63
7.0e-04
0.920 ±0.002
0.205 ±0.011
0.021 ±0.027
1.1e-01
0.427 ±0.029
0.474 ±0.019
2.2e-80
1.1e-55
1.3e-64
1.4e-05
0.908 ±0.004
0.680 ±0.003
0.971 ±0.011
5.2e-02
0.737 ±0.047
0.623 ±0.059
1.1e-73
3.0e-46
9.6e-63
7.2e-05
0.925 ±0.004
0.680 ±0.002
0.967 ±0.014
5.9e-02
0.470 ±0.014
0.607 ±0.012
4.4e-80
8.3e-51
2.2e-57
1.6e-05
0.937 ±0.001
0.686 ±0.010
0.969 ±0.011
2.9e-02
0.770 ±0.002
0.711 ±0.036
8.6e-76
5.4e-42
1.4e-65
2.3e-06
0.906 ±0.001
0.594 ±0.010
0.800 ±0.008
3.8e-02
0.630 ±0.018
0.649 ±0.016
1.5e-87
7.5e-50
4.6e-70
3.2e-04
Recon R² (norm)
Recon R² (orig)
Max-val R² (pooled)
MMD² ↓
Density@5 ↑
Coverage@5 ↑
RQA-DET KS p ↑
RQA-L_mean KS p ↑
RQA-LAM KS p ↑
λ1 KS p ↑